In 1584, Scribonius was the first to describe dark coloured urine, looking at the caseof a child, which excreted urine which turned "black as ink" on exposure to air. In 1859, Boedekerused, for the first time, the term alkaptonuria. It is a rare congenital metabolic abnormality where urine contains large quantities of homogentisic acid. Homogentisic acid (2,5, dihydroxyphenylacetic acid), is the key intermediate in the metabolism of phenyalanine and tyrosine.. A block in the conversion of homogentisic acid to maleyl aceto acetic acid is due to deficiency of the enzyme homogentisic oxidase. Consequently, homogentisic acid (called Alkapton) is excreted in the urine and the defect is called Alkaptonuria. The urine becomes dark brown or black on standing through oxidation by air.

7.5.1 Causes

Alkaptonuria is characterized by the deficiency of homogentisate 1, 2 dioxidasewhich catalyses the conversion of homogentisic acid to Maleylacetoacetic acid. Ochronosis,a dark pigment in connective tissue,is characteristic of this disorder. Ochronotic pigment may be sedimented in any of the outer structures of the eye: sclera, cornea, conjunctiva. The pigment can be seen in the nasal and temporal aspects of the sclera (Fig.7.8).



Figure. 7.10 Symptoms of alkaptonuria

Alkaptonuria does not show any ill effects in early life but it leads to degenerative arthritis in old age. This is because of the crystallization of the homogentisic acid derivatives in cartilages of ears and other exposed places. This results in generalized pigmentation of connective tissues and deposition in joints leading to arthritis.

7.5.2 Symptoms

It has a very low prevalence (1:100,000-250,000) in most ethnic groups. Two countries, Slovakia and Dominican republic, exhibit an increased incidence of this disorder. The first clinical sign is the appearance of dark spots in nappies of the affected children, due to darkening of urine owing to oxidation of the HGA. Till now no treatment modalities are known to treat this disease. The best way is to reduce the intake of tyrosine and

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